

LESSON 8.7

INTRODUCTION TO SYSTEMS OF TWO LINEAR EQUATIONS



LESSON INTRODUCTION

MA.8.AR.4.1 Given a system of two linear equations and specified set of possible solutions, determine which ordered pairs satisfy the system of linear equations.

MA.8.AR.4.2 Given a system of two linear equations represented graphically on the same coordinate plane, determine whether there is one solution, no solution, or infinitely many solutions.

LEARNING TARGETS

- I can relate equations with variables on both sides to two variable linear equations.
- I can relate the solution to an equation with variables on both sides to the point of intersection on a graph.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two variable linear equations algebraically or graphically.

ESSENTIAL QUESTIONS

- What is a system of equations?
- What does it mean to be a solution to a system of equations?
- How many solutions do systems of equations have?

LESSON NARRATIVE

In this lesson, students explore what a system of equations is for the first time. Students work with partners to explore the definition and relate it to a system with one solution, both algebraically and graphically. Students are then led through further exploration of a system with no solution and one with infinitely many solutions. While students are not expected to solve systems of equations algebraically in Grade 8, this lesson includes solving equations where the two linear expressions are set equal to each other to find when they are the same. These components are intentionally included in a scaffolded exploration or guided components of the lesson where the teacher models and explains.

COMPONENT SUMMARY

8.7.1 WARM-UP (~5 MIN.)

Write a large number in scientific notation

8.7.2 EXPLORATION (15-20 MIN.)

Explore a system of linear equations

8.7.3 GUIDED INSTRUCTION (15-20 MIN.)

Solve systems of linear equations with no solutions and infinitely many solutions

8.7.4 WRAP-UP (~5 MIN.)

Indicate level of understanding of systems of linear equations

RESOURCES

GLOSSARY

Key Terms:

- System of equations

Additional Terms: N/A

MATERIALS

- (Optional) Scientific calculator

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may continue to struggle with solving one-variable equations.
- Students may struggle to make connections between the equations in a system and the number of solutions it has.

THINGS TO CONSIDER

- This lesson consistently uses slope-intercept form so students can focus more on the idea of finding solutions and discovering the patterns.
- Consider opportunities to spiral content from linear relationships learned in Unit 7 as students relate the equations in systems to the number of solutions.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

8.7.2 Exploration positions students to analyze a graph of a system of linear equations. Consider applying the “Notice and Wonder” routine to this part of the lesson, positioning students to first consider what makes sense to them from the graph of a system of linear equations before delving deeper into this new concept.

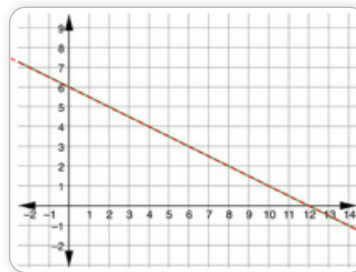
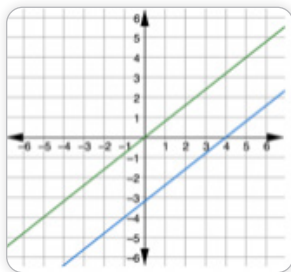
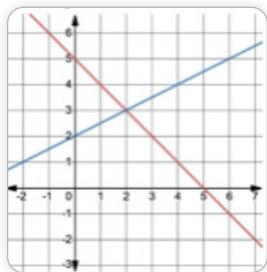
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

Students will be positioned to look for patterns and structures when determining the number of solutions to linear systems of equations. In the 8.7.3 Guided Instruction, students use algebraic thinking and graphs of particular systems to predict and determine the number of solutions of a system of linear equations.

DIFFERENTIATE INSTRUCTION

Utilize the structure of the lesson by positioning students to focus on the graphs of the systems of linear equations referenced during both the Exploration and the Guided Instruction. This will assist students who need a visual of the graph to assist with making sense of the relationships between the number of solutions to a system and the graph of the system.



8.7.1 WARM-UP: DIAMOND IN THE SKY

TEACHER MOVES

Monitor students as they respond and note students to share during whole-group discussion.

DIFFERENTIATE INSTRUCTION

Consider challenging students to explain their answers either verbally or through writing. Having to vocalize or write out the thought process and logic behind the answers positions students to think deeply about how to arrive at the answer.



8.7.1 Warm-Up: Diamond in the Sky

In 2004, the Harvard Smithsonian Center for Astrophysics discovered the biggest diamond in the galaxy. This diamond once known as star BPM 37093, is now known as “Lucy” from a song by the Beatles. Lucy is 50 light years away from Earth and is approximately 2,500 miles across with a weight of approximately 10 billion-trillion-trillion carats.

- Write “10 billion-trillion-trillion” in scientific notation.
 1×10^{34}

- Rate your confidence in your answer to the previous question.

Answers will vary.

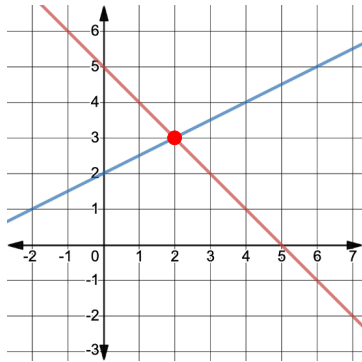
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8.7.2 Exploration: Systems of Linear Equations

Work with your partner to complete the following.

- The equations $y = -x + 5$ and $y = \frac{1}{2}x + 2$ are shown on the coordinate grid.



- What do you notice about the graph?
Answers will vary. There are two lines that intersect each other. One of the lines is red, one is blue. One of the lines has a positive slope and one has a negative slope.

A **system of equations** is a set of two or more equations. Each equation contains two or more variables. The solution is the values for the variables that make all equations in the system true.

8.7.2 EXPLORATION: SYSTEMS OF LINEAR EQUATIONS

TEACHER MOVES

This Exploration is intentionally scaffolded to allow students to engage with their partner rather than being guided through it by the teacher.

Additional scaffolding questions are provided if needed, but this should be a primarily student-directed Exploration.

TEACHER MOVES

Notice and Wonder

Consider applying the language routine “Notice and Wonder” while students analyze the graph of the system of equations. Encourage students to make conjectures about the meanings of the parts of the equations, the graphs of both lines, and the intersection of the two lines.

Students should not be pushed toward using language like “system of equations” or “solution.” They will likely make connections between which line is which based on their work in Unit 7 and may wonder why both lines are shown on the same coordinate grid. This prompt is meant to give them time to consider the graph and its features before delving deeper into new concepts.

TEACHER MOVES

This is the first time students are being introduced to a system of equations. They are not expected to recognize the significance of the point of intersection just yet. While some students may make a connection to this point as the solution after reading the definition, others will not. It’s not the expectation that students are at this point yet. Continue to let them explore this system with their partner.

8.7.2 EXPLORATION: SYSTEMS OF LINEAR EQUATIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems for additional scaffolding for questions 1b and 1d.

- This is a system of equations because it includes _____ equations.
- The equations each contain _____.
- When values make an equation true, that means _____. So for a system of equations, that means _____.

- b. Use the definition to explain why the graph is a system of equations.

Answers will vary. This is a system of equations because it is a set of two or more equations and each equation contains at least two variables. In this case, the variable used are x and y .

- c. Discuss with your partner what the quote from the definition, "The solution is the values for the variables that make all equations in the system true" means. Summarize your answer.

Answers will vary. This means that any (x, y) coordinate pair that satisfies both equations when x - and y -values are substituted is a solution to the system. These would also be any point that is on both lines when graphed.

To find the values for the variables that make both $y = -x + 5$ and $y = \frac{1}{2}x + 2$ true, the equations can be set equal to each other.

Since $y = -x + 5$ and $y = \frac{1}{2}x + 2$, then $-x + 5 = \frac{1}{2}x + 2$.

- d. Solve the equation for x .

$$\begin{aligned} -x + 5 &= \frac{1}{2}x + 2 \\ 5 &= 1\frac{1}{2}x + 2 \\ 3 &= 1\frac{1}{2}x \\ 2 &= x \end{aligned}$$

TEACHER MOVES

Question 1i is a discussion opportunity. Students do not need to write an explanation here.

Listen as students discuss their thinking in response to this prompt to determine whether students are making connections. Misconceptions can be addressed during a whole-class debrief after students have finished question 2.

TEACHER MOVES

Facilitate a brief whole-class discussion before moving to the next component. Highlight student thinking by calling on students to share their thought processes regarding what a system of equations is and what it means to be a solution to a system.

- e. The solution only gives the x -value. Decide who will be Partner A and who will be Partner B. Each partner uses the provided work to substitute the value of x into one of the equations to find y .

Partner A	Partner B
$y = -x + 5$	$y = \frac{1}{2}x + 2$
$y = -(2) + 5$	$y = (\frac{1}{2} \cdot 2) + 2$
$y = -2 + 5$	$y = 1 + 2$
$y = 3$	$y = 3$

- f. Verify the y -value with your partner.
- g. Use the x - and y -values to create an ordered pair.
 $(x, y) = (2, 3)$
- h. Plot your point on the graph.
Shown on graph.
- i. Discuss with your partner how to use the graph to find the solution of the system.

The definition of systems of equations states, "The solution is the values for the variables that make all equations in the system true."

2. Complete the statement to explain how the graph of the system shows the values of the variables that make both $y = -x + 5$ and $y = \frac{1}{2}x + 2$ true.

The solution of a system of equations is the

- point of intersection.
- slope.
- y -intercept.

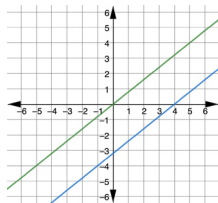


8.7.3 Guided Instruction: Solving Linear Systems

1. A system of equations is shown.

$$\begin{cases} y = 0.8x \\ y = 0.8x - 3.2 \end{cases}$$

The system of equations is also shown on the coordinate grid.



- a. Solve for x .

$$0.8x = 0.8x - 3.2$$

$$0.8x - 0.8x = 0.8x - 3.2 - 0.8x$$

$$0 \neq -3.2$$

There is no solution.

- b. Discuss with your partner how the solution to $0.8x = 0.8x - 3.2$ and the graph of the system are related. Summarize your discussion.

Answers will vary. There is no solution to the equation $0.8x = 0.8x - 3.2$. The graph of the system also shows no point of intersection. Therefore, there is no solution to the system either. The lines are parallel lines.

2. Consider the following system of linear equations.

$$\begin{cases} y = -0.5x + 6 \\ y = 6 - \frac{1}{2}x \end{cases}$$

- a. Solve the system.

$$-0.5x + 6 = 6 - \frac{1}{2}x$$

$$-0.5x + 6 = -\frac{1}{2}x + 6$$

The two linear expressions are equivalent expressions. Therefore, there are infinitely many solutions to the equation when they are set equal to each other. Therefore, the system has infinitely many solutions.

8.7.3 GUIDED INSTRUCTION: SOLVING LINEAR SYSTEMS

TEACHER MOVES

Explain why the linear expressions are set equal to each other to solve the system. Similar to when solving real-world problems in Lesson 5, setting the linear expressions equal to each other allows us to find when the y -values of both equations will be the same.

Model how to do this for question 1a, making explicit connections to what students learned about the number of solutions to an equation based on its structure.

TEACHER MOVES

Think-Pair-Share

For question 1b, provide students with the opportunity to independently read, process, and answer the prompt in question 1b before discussing with a partner. Think-Pair-Share gives students three opportunities to make sense of information. Debrief this prompt with the whole group to ensure students are taking away the intended conclusions.

ENRICHMENT

Model question 2a using a Think-Aloud:

- How can I solve this system? (Set the linear expressions equal to each other and solve the equation.)
- "The expression $6 - \frac{1}{2}x$ can be rewritten as $-0.5x + 6$. These two expressions are equivalent. That means there are infinitely many solutions to this equation."

8.7.3 GUIDED INSTRUCTION: SOLVING LINEAR SYSTEMS (CONTINUED)

TEACHER MOVES

Students are not just copying their classmates' predictions, but instead critically listening and taking notes. Students record their classmates' predictions to consider with their own as they continue making sense of the observations they have made.

TEACHER MOVES

Guide students through questions 2d-2g to ensure they are making the connections after exploring their ideas in questions 2b-2c.

TEACHER MOVES

MA.K12.MTR.5.1: Patterns and Structure

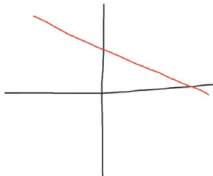
Continue encouraging students to determine the relationship between the number of solutions and the types of equations given in the system. Probe students to look for patterns or characteristics in each set of equations or graphs to position students to make connections.

DIFFERENTIATE INSTRUCTION

If time allows, consider positioning students to summarize their learning regarding the characteristics of systems of linear equations and their number of solutions by creating a graphic organizer or foldable. Make intentional connections to prior learning regarding the number of solutions for a single linear equation.

- b. Discuss with your partner what you noticed while solving the system. Predict what the graph of the system of equations would look like. Make a sketch of your prediction.

Answers will vary. I notice the equations are equivalent so both lines should be the same graph. The graph should appear to have one line.

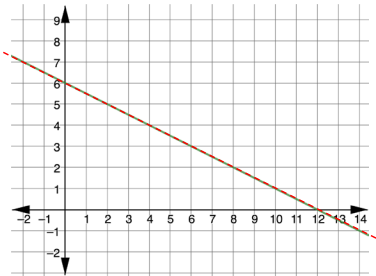


- c. Ask a classmate what they noticed and their prediction. Record their observation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Answers will vary.

Notice	My Summary of Their Reason	Initials
Predicted Graph	My Summary of Their Reason	Initials

- d. The graph of $y = 6 - \frac{1}{2}x$ is given.



- e. Graph $y = -0.5x + 6$ on the same coordinate plane.
See red dashed line above.
- f. How many x - and y -value pairs are the same for these two lines?
They are all the same. All the points on both lines are solutions for this system of equations.

- g. How many solutions does this system have?
It has an infinite number of solutions.



8.7.4 Wrap-Up: Growth Mindset

1. Consider which sign best represents the road you are on.
Answers will vary.
- a. Choose the question and sign that best describes your understanding of today's lesson by putting an asterisk (*) next to the sign that matches.

Are you confused?	Are you in the school zone?	Are you ready for the highway?
		

- b. Briefly explain why you feel this way.
Answers will vary.

8.7.4 WRAP-UP: GROWTH MINDSET

TEACHER MOVES

Using the student explanations, consider making a list of the what students are still confused about on the board or poster to come back to and address later.

TEACHER MOVES

Consider pairing students to discuss their reflections to see if peers can answer any of their questions.